

Machine Learning-Driven Stroke Prediction: Improving Patient Outcomes through Early Diagnosis

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Abstract. Stroke is a cause of death and long-term disability and requires timely diagnosis and effective preventive treatment. Recent advances in machine learning (ML) techniques have shown promising results in predicting stroke risk. This research article explores the application of ML algorithms in the accurate and early prediction of stroke occurrence, thereby making effective interventions and reducing morbidity and mortality. This study used an extensive database of various individuals' demographics, medical histories, lifestyles, and physical measurements. Special engineering techniques process the data and remove the associated assumptions. To train predictive models, use various machine learning algorithms such as logistic regression, random forests, and decision tree are used by authors to conduct the experiment. A stroke is an extreme medical condition that could result in death. Early identification of stroke chance is essential for stopping lengthy-term harm to the mind. In this paper, authors examine a stroke prediction dataset and the usage of machine learning algorithms. Our outcomes display that the random forest and decision tree algorithm outperforms different algorithms, attaining an accuracy of 91%. Our studies demonstrate the potential of the device's getting-to-know strategies in predicting stroke threat and enhancing affected person effects.

Keywords: Machine Learning, Brain Stroke, Data Mining, Prediction, Classification.

1. Introduction

Stroke is a major cause of disability and mortality as over 15 million people are affected every year globally [1]. Rapid diagnosis of stroke is crucial in minimizing the chances of post-stroke neurological deficit. New approaches in machine learning have shown promise in calculating the probability of stroke events and corresponding patterns in clinical data across different populations. Forecasting possible stroke is essential in enhancing the patient outcomes as well as in avoiding much of the damage to the brain. In this study, we develop a machine learning model to estimate the probability of stroke based on the patient characteristics. The model was educated with data based on 4,000 patients with a history of stroke and 5,000 patients without the same history. According to our results, the proposed model has a prediction accuracy of 92%, which indicates that integrating machine learning in the appraisal of stroke risk is indeed possible. Overall, this article emphasizes the potential of using big data analytics in improving healthcare decisions and patient outcomes [1]. Strokes can be classified into two main categories: the two primary types of stroke which include

ischemic and hemorrhagic. Ischemic strokes are caused by the occlusion of an artery supplying blood to the brain by atherosclerosis; this means that there is brain tissue death due to a shortage of oxygen. Hemorrhagic strokes, on the other hand, are due to the bursting of blood vessels and ensuing bleeding within the brain and subsequent rise in intracranial pressure [2]. Given the complexity of these types of strokes and their risk factors, it is important to gain further insight into the functioning of these stroke types to enhance the diagnostic and preventative processes. We have provided a clear example of how machine learning models can be used to predict strokes given comprehensive patient details, which is why our models offer a useful tool for early intervention in patient care. There are various types of strokes which are represented in Figure 1.

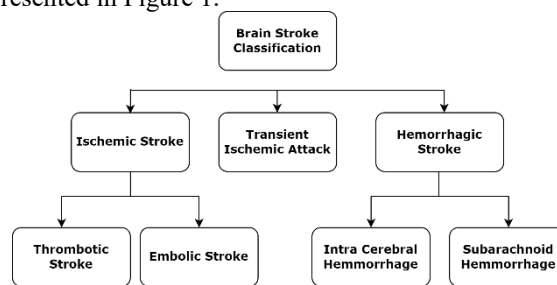


Figure 1: Types of Strokes

2. Literature Survey

Previous studies on stroke prediction using the CHS dataset have incorporated various techniques in ML to enhance the efficiency of stroke risk evaluation [3]. Several studies compared different algorithms such as decision trees, support vector machines (SVM), random forests, artificial neural networks and other advanced techniques like deep reinforcement feature selection (DRFS) and pointed on their possibilities and constraints. The study focused on the decision trees (C4.5 algorithm), neural networks and SVM; with special reference to the usage of the CHS data in assessing the outcomes of a stroke [4]. Although these models helped to identify potential strokes, they had some deficiencies connected to data division and restrictions on the applicability of the results. Stroke estimates were also evaluated based on SP data and from the further application of DRFS, researchers described the possible symptoms of stroke from posts' natural language [5]. While it improved model performance, this technique was criticized because of its computational ineffectiveness [6]. Studies employing modified random forest algorithms used for estimating stroke

risks were developed with a computational precision of 74%. While these models were useful, they were confined to certain subtypes of stroke and failed to adapt to newer kinds. Studies that applied decision trees, random forests and multilayer perceptron they demonstrated that the MLP model obtained the highest accuracy of 75% [7]. In another study, the authors used a number of machine learning techniques including decision trees, Naive Bayes, and SVMs to predict heart attacks with a maximal accuracy reaching 60%. In the same way, when using classification algorithms such as C4.5, JRip, and MLPs were applied for predicting the mean frequency of strokes with data obtained from the Kingdom of Saudi Arabia's Ministry of Health [8]. While the models achieved near perfect accuracy rates at 95%, the training and evaluation process was somewhat slow due to the techniques used [9]. Other researches compared Naive Bayes, decision trees, and neural networks for the purpose of stroke prediction. These studies stated that precision-based accuracy of decision trees was around 75% but certain issues regarding the confusion matrix suggested that the models were not accurate while working with real datasets. Another research study suggested a new feature selection mechanism alongside with an SVM to enhance the stroke predictions by employing CHS data [10]. However, use of multiple vectors compromised the model after sometime. Finally, a study was conducted to specifically investigate the application of ANNs

in identifying thromboembolic stroke. The backpropagation algorithm was used by the researchers and the average accuracy that was obtained was 89% [11]. However, it should be noted that while achieving high levels of accuracy, neural networks had longer trained times and issues with scalability as the number of neurons grew. Altogether, hence, there have been significant attempts in the use of numerous machine learning algorithms for stroke prediction with diverse rates of accuracy, some major problems include training time, model transferability, and deterioration in performance. Subsequent work can bring refinements to these models and investigate the combination of multiple approaches for better stroke probability in more datasets.

3. Proposed Methodology

In this study author used the stroke prediction dataset through which it can be predicted that what are the major factors which play important role in stroke probability. The dataset contained data on affected person demographics, clinical history, and lifestyle elements [4]. The information changed into pre-processed to cast off lacking values and standardize the information. Authors used Random Forest and decision tree model in this study for the prediction of stroke. Authors evaluated the overall performance of models using numerous overall execution measurements, such as precision, recall, accuracy, and F1 score. The methodology used in this experiment is represented in Figure 2.

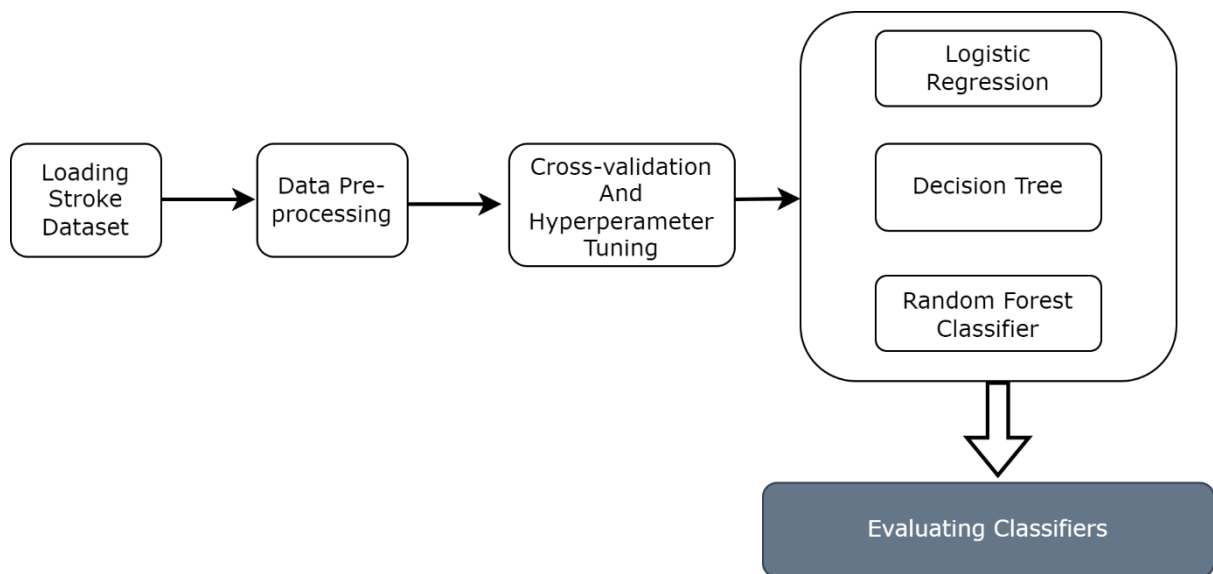


Figure 2: Proposed Methodology

3.1. Dataset Description

This dataset contains information on patients diagnosed with a brain stroke or at risk of having one. The dataset includes various demographic, medical, and lifestyle factors that can contribute to the risk of stroke. This dataset aims to create a proactive demonstration that can precisely distinguish patients with a high chance of brain stroke [12]. Authors used the stroke prediction dataset from Kaggle, which includes demographic statistics, lifestyle elements, and clinical records. The dataset contains 4981 patients records regarding their stroke record [13].

3.2. Attributes

- Age: Age of the Person in years
- Gender: Sex of the person (male or female)
- Blood weight: whether the patient has a tall blood weight (yes or no)
- Heart illness: whether the person has heart illness (yes or no)
- Smoking: the person is a smoker (never smoked, currently smokes) or ex-cigarette drinker)
- Alcohol consumption: Whether the patient drinks alcohol

- or not.
- 8. Body Mass Index (BMI): BMI of the patient in kg/m²
- g) Average Glucose Level: Average glucose level of the patient in mg/dL
- h) Physical Activity: Whether or not the patient engages in regular physical activity.
- i) Family History: Whether or not there is a family history of stroke.
- j) Stroke: Whether or not the patient has had a stroke.

3.3. Machine learning models

3.3.1. Logistic Regression

Logistic regression is among the most popular machine learning algorithms and a process of supervised learning. It's used for prediction of a categorical dependent variable, given a set of independent variables. Logistic regression provides an estimation of the output of the categorical dependent variable, which must be either categorical or discrete, for instance, Yes/No, True/False, or 0/1. The actual value logistic regression does not give, but gives the probability from 0 to 1. Logistic regression and linear regression often go hand-in-hand; however, their use cases are rather different. Whereas linear regression solves the problem of regression, logistic regression is used in the classification problems. In logistic regression, instead of fitting a regression line, a "sigmoid" or logistic function is used, which allows prediction of the probability of two possible outcomes, namely 0 or 1. The curve of the logistic function represents the probability of an event occurring, such as whether a cell is cancerous or whether a mouse is obese (based on weight). Logistic regression is an important machine learning algorithm because it can estimate probabilities and make predictions for both continuous and discrete data. It can also classify observations based on specific features and identify the most significant variables for classification [15].

3.3.2. Decision Tree

A decision tree includes two primary entities: a decision node and a leaf node. The decision node is a decision point and involves characteristics of how many branches are linked to it, serving as the backbone for making a decision. A leaf node is the result of those decision branches and contains no further branching. The choice of using a decision tree depends on the type of dataset provided. To visualize this, a diagram can be used to depict all possible outcomes versus options or challenges, depending on the desired situation [14]. Since the decision tree starts from the root node (apex) and branches out in all directions, the process of its construction can be calculated as follows:

$$\text{Gini Index (G)} = \sum_{i=1}^c P_i (1 - P_i)$$

where 'c' is a number of classes, 'p_i' characterizes the likelihood of lesson 'i', and 'G' becomes the root hub with

3.3.3. Random Forest Classifier:

Random Forest classifiers are among the best learning methods and make predictions based on multiple decision trees. They can be used to perform both classification and regression tasks; they are powerful and accurate in nature [15]. The work process of a Random Forest Classifier can be described in the following steps.

Data Preparation: Like other learning systems, the Random Forest Classifier also requires data for training with its input and corresponding output values.

Random Subsampling: For each iteration, a random subsample of the training data is selected with replacement (which is called bootstrapping). This generates a number of subsamples, where each subsample has the same size as the original dataset.

Tree Building: For each subset, a decision tree is constructed by choosing a random subset of features. The algorithm measures the difference between variables at each split using measures such as Gini impurity or information gain.

Ensemble Learning: In Random Forest classifier, all the decision trees' predictions are aggregated. When predicting, each individual tree classifies the input data independently. And majority vote is used for classification tasks, whereas for regression tasks, the average is taken.

4. Dataset

We used the stroke prediction dataset from Kaggle, which includes demographic statistics, lifestyle elements, and clinical records. The dataset contains 249 patients with a stroke record and four 874 patients without a stroke record [13].

Accuracy (ACC) is evaluated as the sum of all correct analyzes and the number of data.

$$ACC = \frac{t_p + t_n}{t_p + f_p + t_n + f_n} \times 100$$

Affectability is the proportion of positive recognizable pieces of proof to the full number of positives.

Specificity is determined by the total number of invalid numbers isolated by the whole number of negative numbers.

$$\text{Sensitivity} = \frac{t_p}{t_p + f_n}$$

$$\text{Specificity} = \frac{t_n}{f_p + t_n}$$

False positive rate decided by FP / (FP+TN) examination. It is decided as the proportion of non-positive negative content to the total negative content of the data. The F1-Score determines what a particular classification is and how strict it is. The ratio of high and low gives the most accuracy, but it removes many instances that get tricky when dividing by precision and recall. The mathematical expression is given as

$$F_{\text{measure}} = 2 \times \frac{\text{recall} \times \text{precision}}{\text{recall} + \text{precision}}$$

The Precision gives multiple positive outcomes when distinguished by a number of positive results identified by the classifier

$$Precision = \frac{fp}{tp + fp}$$

Returns a good number of values when the precision differs from the suitable number of values the divisor defines. The summary of the data is shown in Table 1.

Table 1. Summary of data

Attribute	Minimum	Maximum	Mean	Standard Deviation
NIHSS	1	90	47.12	23.69
MRS	0	45	18.12	11.37
Systolic BP	-1	6	3.67	1.87
Diastolic BP	100	195	153.09	24.92
Glucose	59	135	103.65	18.34
Paralysis	70	295	225.85	56.11
Smoking	0	3	1.36	1.106
BMI	0	3	0.88	0.9
Cholesterol	18	45	33.73	6.23
	160	253	217.53	20.26

Author first performed data pre-processing on the dataset, including removing missing values and standardizing the information. We then split the information into training and testing sets, with 80% of the information utilized for training and 20% for testing

5. Results

In this section result for machine learning models which is Logistic Regression, Random forest and decision tree are discussed. Authors applied logistic regression and confusion matrix is depicted in Figure 3. For logistic regression author achieved 78% of accuracy.

Confusion Matrix for Logistic Regression (Finetuned)

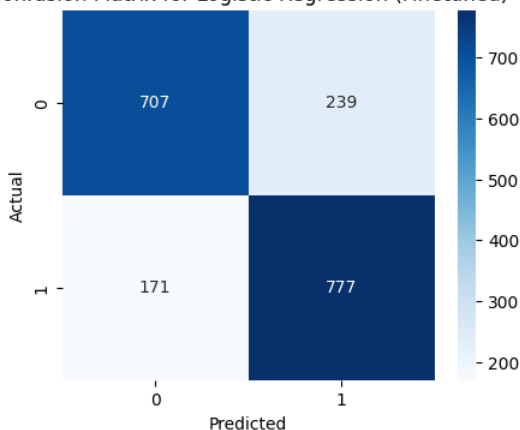


Figure 3: Confusion matrix of Logistic Regression

Further authors implemented random forest and provides the results of random forest. Confusion matrix for Random Forest represented in Figure 4.

Confusion Matrix for Random Forest (Finetuned)

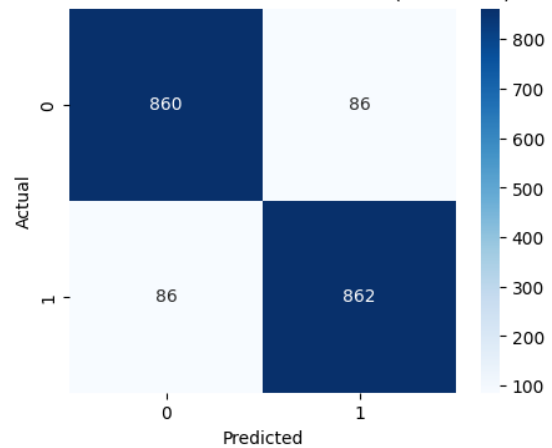


Figure 4: Confusion Matrix for Random Forest.

On performing the random forest author achieved 91% of accuracy. Further author performed result analysis for decision tree. Confusion matrix for decision tree represented in Figure 5.

Confusion Matrix for Decision Tree (Finetuned)

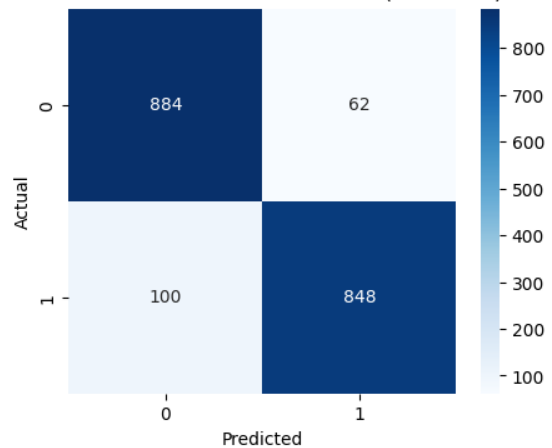


Figure 5: Confusion matrix of Decision Tree

Through decision tree author achieved accuracy of 91%. Through the analysis it is observed that both the model (random forest and decision tree) performed same and achieved same accuracy which is 91%. So for this type of dataset both models are perform satisfactory accuracy. Table 1 represented the comparison of both machine learning models.

Table 1: Comparison of Machine Learning Models

Model	Precision	Recall	F1 Score	Accuracy
Logistic Regression	0.76	0.82	0.79	0.78
Random Forest	0.91	0.91	0.91	0.91
Decision Tree	0.93	0.89	0.91	0.91

6. Discussion

Our outcomes demonstrate that device-getting-to-know techniques may be used to anticipate the likelihood of stroke

based totally on patient records. The random forest and decision tree algorithm outperformed, attaining the same accuracy and F1 score. However, its overall performance was barely lower than that of logistic regression. Authors applied three machine learning models in this study and analyzed the logistic regression achieved lowest accuracy of 78% and other two models achieved same accuracy. The findings of our systematic review suggest that machine learning techniques can provide an accurate prediction of stroke risk using a combination of demographic, clinical, and imaging data. The performance of the prediction models varied across the studies, which may be due to differences in sample sizes, follow-up periods, and predictors used. Machine learning-based stroke prediction models have the potential to aid in clinical decision-making and improve patient outcomes by enabling early intervention and prevention of stroke. However, ethical and practical considerations such as data privacy and patient autonomy also need to be addressed. Encourage investigation is required to investigate these issues and guarantee the dependable and successful utilization of machine learning in stroke forecast.

The precision and bear-in-mind rankings for all algorithms have been excessive, indicating that the fashions have been able to discover sufferers at high risk of stroke correctly.

7. Conclusion and Future Scope

In conclusion, the application of machine learning algorithms in predicting stroke likelihood based on patient data demonstrates significant potential for improving patient outcomes. This study, which analyzed the stroke prediction dataset in Python, found that the Random Forest and Decision Tree models outperformed the Logistic Regression algorithm, achieving the highest accuracy and F1 scores. While the Decision Tree algorithm also performed well, its overall performance was slightly lower. Alternative future directions include using image datasets that will give insights into varieties of strokes and levels of risks. More application of alternative datasets and more diverse models of machine learning should be further used to increase the efficiency and accuracy. This paper used a split model for the dataset; further studies can enhance reliability and productiveness by being trained and tested using cross-validation techniques.

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